

Questions with Answer Keys

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Q1 - Single Correct

The solution set of $\sqrt{5x-6-x^2} + \frac{\pi}{2} \left(\int_0^x dz \right) > x \int_0^\pi \sin^2 x dx$ is

- (1) R
- (2) $(1, 6)$
- (3) $(-6, 1)$
- (4) $(2, 3)$

Q2 - Single Correct

For $U_n = \int_0^1 x^n (2-x)^n dx$; $V_n = \int_0^1 x^n (1-x)^n dx$, $n \in N$, which of the following statement is true?

- (1) $U_n = 2V_n$
- (2) $U_n = 2^{-n}V_n$
- (3) $U_n = 2^{2n}V_n$
- (4) $U_n = 2^{-2n}V_n$

Q3 - Single Correct

If $\int_0^{f(x)} t^2 dt = x \cos \pi x$, then $f'(9)$ is equal to

- (1) $-1/9$
- (2) $-1/3$
- (3) $1/3$
- (4) non-existent

Q4 - Single Correct

The positive value of a , so that the definite integral $\int_a^{a^2} \frac{dx}{x+\sqrt{x}}$ achieves smallest value, is

- (1) $\tan^2 \frac{\pi}{8}$
- (2) $\tan^2 \frac{3\pi}{8}$
- (3) $\tan^2 \frac{\pi}{12}$
- (4) 0

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Q5 - Single Correct

If $f''(x) = -\frac{1}{x^2} - \pi^2 \sin(\pi x)$; $f'(2) = \pi + \frac{1}{2}$ and $f(1) = 0$, then the value of $f\left(\frac{1}{2}\right)$ is

(1) $\log 2$

(2) 1

(3) $\frac{\pi}{2} - \log 2$

(4) $1 - \log 2$

Q6 - Single Correct

Let $I(a) = \int_0^\pi \left(\frac{x}{a} + a \sin x\right)^2 dx$; $a \in \mathbb{R}$. Then, the value of a for which $I(a)$ attains its minimum value, is

(1) $\sqrt{\pi\sqrt{\frac{2}{3}}}$

(2) $\sqrt{\pi\sqrt{\frac{3}{2}}}$

(3) $\sqrt{\frac{\pi}{16}}$

(4) $\sqrt{\frac{\pi}{13}}$

Q7 - Single Correct

$\int_{\pi/2}^{3\pi/2} e^{-x/2} \cdot \frac{\sqrt{1-\sin x}}{1+\cos x} dx$ is equal to

(1) $\left(2e^{-\pi/3} - \sqrt{2}e^{-\pi/4}\right)$

(2) $\left(e^{-\pi/3} - \sqrt{2}e^{-\pi/4}\right)$

(3) $\left(e^{-\pi/2} \frac{2}{\sqrt{3}} - e^{-\pi/4} \sqrt{2}\right)$

(4) None of these

Q8 - Single Correct

Let $f(x) = \int_2^x \frac{dt}{\sqrt{1+t^4}}$ and g be the inverse of f , then the value of $g'(0)$ is

(1) 1

(2) 17

(3) $\sqrt{17}$

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(4) None of these

Q9 - Single Correct

Let $a_n = \int_0^{\pi/2} (1 - \sin t)^n \sin 2t dt$, then $\lim_{n \rightarrow \infty} \sum_{n=1}^n \frac{a_n}{n}$ is equal to

(1) $1/2$

(2) 1

(3) $4/3$

(4) $3/2$

Q10 - Single Correct

Let $C_n = \int_{1/(n+1)}^{1/n} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)} dx$, then $\lim_{n \rightarrow \infty} n^2 \cdot C_n$ is equal to

(1) 1

(2) 0

(3) -1

(4) $\frac{1}{2}$

Q11 - Single Correct

If $f(x) = e^{g(x)}$ and $g(x) = \int_2^x \frac{t}{1+t^4} dt$, then $f'(2)$ is equal to

(1) $2/17$

(2) 0

(3) 1

(4) Cannot be determined

Q12 - Single Correct

If $f(x) = \sin x + \int_0^x f'(t) (2 \sin t - \sin^2 t) dt$, then $f(x)$ is equal to

(1) $\frac{x}{1-\sin x}$

(2) $\frac{\sin x}{1-\sin x}$

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(3) $\frac{1-\cos x}{\cos x}$

(4) $\frac{\tan x}{1-\sin x}$

Q13 - Single Correct

If $g_n(x) = x^{2n+1} + a_n x + b_n (n \in N)$ satisfies the equation $\int_{-1}^1 (px + q)g_n(x)dx = 0$ for all linear function $(px + q)$, then

(1) $a_n = b_n = 0$

(2) $b_n = 0, a_n = -\frac{3}{2n+3}$

(3) $a_n = 0, b_n = -\frac{3}{2n+3}$

(4) $a_n = b_n = n$

Q14 - Single Correct

Let $f(x) = \int_{-1}^x e^{t^2} dt$ and $h(x) = f[1 + g(x)]$, where $g(x)$ is defined for all x , $g'(x)$ exists for all x , and $g(x) < 0$ for $x > 0$. If $h'(1) = e$ and $g'(1) = 1$, then the possible value which $g(1)$ can take, is

(1) 0

(2) -1

(3) -2

(4) -4

Q15 - Single Correct

The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}$, $x > 0$, is

(1) $x \tan^{-1} x$

(2) $\tan^{-1} x$

(3) $\frac{\tan^{-1} x}{x}$

(4) $\frac{\tan^{-1} x}{x^2}$

Q16 - Multiple Correct

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Let $a > 0$ and $f(x)$ be monotonically increasing such that $f(0) = 0$ and $f(a) = b$, then

$\int_0^a f(x)dx + \int_0^b f^{-1}(x)dx$ is equal to

(1) $a + b$

(2) $ab + b$

(3) $ab + a$

(4) ab

Q17 - Multiple Correct

Let $f : R^+ \rightarrow R$ be a differentiable function with $f(1) = 3$ and satisfying

$$\int_1^{xy} f(t)dt = y \int_1^x f(t)dt + x \int_1^y f(t)dt, \forall x, y \in R^+, \text{ then}$$

(1) $f(x) = 3 \log_e e(x)$

(2) $f(x) = \log_e e(x)$

(3) $f'(y) = \frac{3}{y}$

(4) $f'(x) = \frac{1}{x}$

Q18 - Multiple Correct

If $I = \int_0^{\pi/2} \frac{dx}{\sqrt{1+\sin^3 x}}$, then

(1) $0 < I < 1$

(2) $0 < I < \frac{\pi}{2\sqrt{2}}$

(3) $\frac{\pi}{2\sqrt{2}} \leq I \leq \frac{\pi}{2}$

(4) $I > 2\pi$

Q19 - Multiple Correct

If $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, then

(1) $\int_0^\infty e^{-2x^2} dx = \frac{\sqrt{\pi}}{2\sqrt{2}}$

(2) $\int_0^\infty xe^{-x^2} dx = \frac{1}{2}$

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$$(3) \int_0^\infty x^2 e^{-x^2} dx = \frac{\sqrt{\pi}}{4}$$

$$(4) \int_0^\infty x^2 e^{-x^2} dx = \frac{\pi}{4}$$

Q20 - Multiple Correct

If $\alpha, \beta (\alpha < \beta)$ are two roots of $(6x + 1)x = 1 + \left[\cos \frac{\pi}{4}\right]$. Then, $\int_0^{2\alpha} \sin\left(\frac{\pi[x]}{2}\right) dx + \int_0^{3\beta} \cos(\pi[x]) dx$ (where, $[\cdot]$ denotes greatest integer function), is equal to

$$(1) \alpha + \beta$$

$$(2) 2$$

$$(3) 0$$

$$(4) [2\alpha + 9\beta]$$

Q21 - Multiple Correct

If $\int_1^x \frac{\log t}{1+t} dt$, where $x > 0$, then the value(s) of x satisfying the equation $f(x) + f\left(\frac{1}{x}\right) = 2$ is/are

$$(1) e$$

$$(2) 2$$

$$(3) e^2$$

$$(4) e^{-2}$$

Q22 - Multiple Correct

Let $f(x) = \int_{-1}^1 (1 - |t|) \cos(xt) dt$, then

$$(1) f(0) \text{ is not defined}$$

$$(2) \lim_{x \rightarrow 0} f(x) = 2$$

$$(3) \lim_{x \rightarrow 0} f(x) = 1$$

$$(4) f(x) \text{ is continuous at } x = 0$$

Q23 - Multiple Correct

If $f[f(x)] = 1 - x$ for all $x \in [0, 1]$ and $J = \int_0^1 f(x) dx$, then

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(1) $J = 1/2$

(2) $f(1/4) + f(3/4) = 1$

(3) $\int_0^{\pi/2} \frac{\sin x}{(\sin x + \cos x)^3} dx = J$

(4) $f(1/3) \cdot f(2/3) = 1$

Q24 - Multiple Correct

If $f(x) + \int_0^x g(t)dt = \sin x(\cos x - \sin x)$ and $[f'(x)]^2 + [g(x)]^2 = 1$, then $f(x)$ and $g(x)$ respectively can be

(1) $\frac{1}{2}\sin 2x, \sin 2x$

(2) $\frac{\cos 2x}{2}, \cos 2x$

(3) $\frac{1}{2}\sin 2x, -\sin 2x$

(4) $-\sin^2 x, \cos 2x$

Q25 - Multiple Correct

The value of $\int_{-2008}^{2008} \frac{f'(x) + f'(-x)}{(2008)^x + 1} dx$ is

(1) 0

(2) $f(2008) - f(-2008)$

(3) $f(2008) + f(-2008)$

(4) None of the above

Q26 - Multiple Correct

If $I = \int_0^{1/2} \frac{dx}{\sqrt{1-x^{2n}}}$, $n \in N$, then

(1) $[I] = 1$

(2) $0 < I < 1$

(3) $\frac{1}{2} \leq I \leq \frac{\pi}{6}$

(4) $0 < I \leq \frac{1}{2}$

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Q27 - Multiple Correct

If $f(4-x) = f(4+x)$ and $f(8-x) = f(x+8)$ and $f(x)$ is function for which $\int_0^8 f(x)dx = 5$, then $\int_0^{200} f(x)dx$ is equal to

- (1) 100
- (2) 25
- (3) 125
- (4) 75

Q28 - Multiple Correct

$\int_{19}^{37} [\{x\}^2 + 3 \sin(2\pi x)] dx$, where $\{\cdot\}$ denotes fractional part, is equal to

- (1) 0
- (2) 6
- (3) 9
- (4) 10

Q29 - Multiple Correct

The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is (are)

- (1) $\frac{22}{7} - \pi$
- (2) $\frac{2}{105}$
- (3) 0
- (4) $\frac{71}{5} - \frac{3\pi}{2}$

Q30 - Multiple Correct

Let $S_n = \sum_{k=0}^n \frac{n}{n^2+kn+k^2}$ and $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2+kn+k^2}$, for $n = 1, 2, 3, \dots$, then

- (1) $S_n < \frac{\pi}{3\sqrt{3}}$
- (2) $S_n > \frac{\pi}{3\sqrt{3}}$

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$$(3) T_n < \frac{\pi}{3\sqrt{3}}$$

$$(4) T_n > \frac{\pi}{3\sqrt{3}}$$

Q31 - Multiple Correct

For $a \in \mathbb{R}$, $a \neq 1$, $\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} \cdot [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$, then a is equal to

$$(1) 5$$

$$(2) 7$$

$$(3) \frac{-15}{2}$$

$$(4) \frac{-17}{2}$$

Q32 - Paragraph 1

Passage I (For Question 32, 33)

Suppose $f(x)$ and $g(x)$ are two continuous functions defined for $0 \leq x \leq 1$. Given, $f(x) = \int_0^1 e^{x+t} f(t) dt$ and $g(x) = \int_0^1 e^{x+t} g(t) dt + x$

The value of $f(1)$ is

$$(1) 0$$

$$(2) 1$$

$$(3) e^{-1}$$

$$(4) e$$

Q33 - Paragraph 1

The value of $g(0) - f(0)$ is

$$(1) \frac{2}{3-e^2}$$

$$(2) \frac{3}{e^2-2}$$

$$(3) \frac{1}{e^2-1}$$

$$(4) 0$$

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Q34 - Paragraph 2

Passage II (For Question 34, 35)

Suppose a and b are positive real numbers such that $ab = 1$. Let for any real parameter t , the distance from the origin to the line $(ae^t)x + (be^{-t})y = 1$ be denoted by $D(t)$.

The value of $I = \int_0^1 \frac{dt}{[D(t)]^2}$ is

(1) $\frac{e^2-1}{2} \left(b^2 + \frac{a^2}{e^2} \right)$

(2) $\frac{e^2+1}{2} \left(a^2 + \frac{b^2}{e^2} \right)$

(3) $\frac{e^2-1}{2} \left(a^2 + \frac{b^2}{e^2} \right)$

(4) $\frac{e^2+1}{2} \left(b^2 + \frac{a^2}{e^2} \right)$

Q35 - Paragraph 2

The value of b at which I is minimum, is

(1) e

(2) $1/e$

(3) $1/\sqrt{e}$

(4) $\sqrt{e}$

Q36 - Paragraph 3

Passage III (For Question 36, 37)

If $S_n = \int_0^{\pi/2} \frac{\sin(2n-1)x}{\sin x} dx$ and $V_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$, where $n \in N$.

$S_{n+1} - S_n$ is equal to

(1) $\frac{1}{n+1}$

(2) $\frac{1}{2n-1}$

(3) $\frac{1}{n+1} - \frac{1}{2n-1}$

(4) 0

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Q37 - Paragraph 3

 $V_{n+1} - V_n$ is equal to

(1) S_n

(2) S_{n+1}

(3) S_{n-1}

(4) $S_{n+1} - S_n$

Q38 - Paragraph 4

Passage IV (For Question 38, 39)Consider the function defined on $[0, 1] \rightarrow R$, $f(x) = \frac{\sin x - x \cos x}{x^2}$, if $x \neq 0$ and $f(0) = 0$ $\int_0^1 f(x) dx$ is equal to

(1) $1 - \sin 1$

(2) $\sin 1 - 1$

(3) $\sin 1$

(4) $-\sin 1$

Q39 - Paragraph 4

 $\lim_{t \rightarrow 0} \frac{1}{t^2} \cdot \int_0^t f(x) dx$ is equal to

(1) $1/3$

(2) $1/6$

(3) $1/12$

(4) $1/24$

Q40 - Integer Type

A function f is continuous on the positive real axis, has the property that for all choices of $x > 0$ and $y > 0$.The integral $\int_x^{xy} f(t) dt$ is independent of x (and therefore depends only on y). If $f(2) = 2$, then $\int_1^e f(t) dt$ is

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Q41 - Integer Type

Let $I = \int_{k\pi}^{(k+1)\pi} \frac{|\sin 2x|}{|\sin x| + |\cos x|} dx$, $k \in N$ and $J = \int_0^{\pi/4} \frac{dx}{\sin x + \cos x}$, then $I + 4J$ is equal to

Q42 - Integer Type

If $f(x)$ is a periodic function with period T , so that $f(x + 13) + f(x + 630) = 0$ and

$\sum_{p=1}^{777} \int_p^{p+T} f(x) dx = \lambda \int_0^T f(x) dx$, then the value of $\left(\frac{\lambda+T}{2011}\right)$ is

Q43 - Integer Type

Let $f: R \rightarrow R$ and $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$ and given that $\int_0^2 f(x) dx = 5$ then the value of $\frac{1}{25} \int_0^{50} f(x) dx$ is

Q44 - Integer Type

If $f(x) = \int_a^x [f(x)]^{-2} dx$, $a \in R$ and $f(10) = 0$, then $[f(x)]^2 \cdot f'(x)$ is equal to

Q45 - Integer Type

$\int_0^x \left(x - [x] - \frac{1}{2}\right) dx = \frac{f(x) \cdot g(x)}{2}$ (where, $[\cdot]$ and $\{\cdot\}$ are greatest integer and fractional part of x), then the value of $|f(x) - g(x)|$ is

Q46 - Integer Type

If $\int_0^1 \frac{x^\alpha - 1}{\log x} dx = \log a + k$, then the value of k is

Q47 - Integer Type

$\lim_{n \rightarrow \infty} \frac{\pi}{6n} \left[\sec^2\left(\frac{\pi}{6n}\right) + \sec^2\left(2 \cdot \frac{\pi}{6n}\right) + \cdots + \sec^2\left((n-1) \cdot \frac{\pi}{6n}\right) + \frac{4}{3} \right]$ has the value $\frac{1}{\sqrt{k}}$ then the value of k is

Q48 - Integer Type

The interval $[0, 4]$ is divided into n equal subintervals by the points $x_0, x_1, x_2, \dots, x_{n-1}, x_n$, where

$0 = x_0 < x_1 < x_2 < \cdots < x_n = 4$. If $\delta x = x_i - x_{i-1}$ for $i = 1, 2, 3, \dots, n$. Then, $\lim_{\delta x \rightarrow 0} \sum_{i=1}^n x_i \cdot \delta x$ is

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equal to

Q49 - Integer Type

If $\int_0^\infty x^{2n+1} \cdot e^{-x^2} dx = \frac{n!}{k}$, then the value of k is

Q50 - Integer Type

If $\int_0^{400\pi} \sqrt{1 - \cos 2x} dx = k \cdot (200\sqrt{2})$, then the value of k is

Q51 - Integer Type

The number of points at which $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$ attains local maximum, is

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Answer Key

Q1 (4)	Q2 (3)	Q3 (1)	Q4 (1)
Q5 (4)	Q6 (1)	Q7 (4)	Q8 (3)
Q9 (1)	Q10 (4)	Q11 (1)	Q12 (2)
Q13 (2)	Q14 (3)	Q15 (3)	Q16 (4)
Q17 (1, 3)	Q18 (3)	Q19 (1, 2, 3)	Q20 (2, 4)
Q21 (3, 4)	Q22 (3, 4)	Q23 (1, 2, 3)	Q24 (3, 4)
Q25 (2)	Q26 (3)	Q27 (3)	Q28 (2)
Q29 (1)	Q30 (1, 4)	Q31 (2, 4)	Q32 (1)
Q33 (1)	Q34 (3)	Q35 (4)	Q36 (4)
Q37 (2)	Q38 (1)	Q39 (2)	Q40 (4)
Q41 (4)	Q42 (1)	Q43 (5)	Q44 (1)
Q45 (1)	Q46 (1)	Q47 (3)	Q48 (8)
Q49 (2)	Q50 (4)	Q51 (2)	